VARDHAN MISTRY

Computer Vision Engineer (M.Sc. Mechatronics)

+49 176 29028526
mistryvardhan@gmail.com
Chemnitz, Germany
vardhan-mistry
Vardhan1303
Germany (No Sponsorship Required)

Professional Experience

Computer Vision Research Engineer — Master's Thesis [Thesis] **Apr 2025 – Dec 2025**
IAV GmbH *Chemnitz, Germany*

- Reduced false positive rate from 4.03% to 0.17% while maintaining Dice up to 0.94
- Designed training strategy combining real, synthetic, and clean data to improve generalisation
- Identified and mitigated domain gap in synthetic-only training (recall drop 0.17–0.39)
- Developed mask-guided GAN-based pipeline (CycleGAN + VAE) for synthetic mud generation
- Benchmarked DeepLabV3+, SegFormer, and PIDNet across multiple training strategies
- Deployed PIDNet via ONNX achieving 7 ms inference for real-time edge execution

Computer Vision Engineer — Internship **Dec 2024 – Feb 2025**
IAV GmbH *Chemnitz, Germany*

- Developed physics-based raindrop simulation improving synthetic data realism for perception models
- Automated RGB + mask dataset generation, reducing manual annotation effort by 100%
- Benchmarked against Stable Diffusion 2.1 using perceptual metrics and human evaluation

Assistant Manager — Machine Vision & Automation **Aug 2022 – Aug 2023**
Adani Group *Mundra, India*

- Developed OpenCV-based defect detection prototypes, reducing manual inspection time by 35%
- Optimized camera calibration and lighting setup, reducing false-reject rates by 12%
- Contributed to industrial automation deployment in a large-scale smelter project within a cross-functional team

Projects

Platooning Autonomous Robots **GitHub**
Skills: *Python, OpenCV, ArUco, Kalman Filter, RPLiDAR, Raspberry Pi*

- **Leader:** Fused RPLiDAR (135°–225°) and camera for lane-following with real-time obstacle halting at 500mm.
- **Follower:** Eliminated LiDAR dependency using ArUco marker vision, achieved <2.5% distance estimation error.

Object Detection & Depth Estimation **GitHub**
Skills: *Python, OpenCV, YOLOv8, KITTI Dataset*

- Combined YOLOv8 with monocular depth estimation achieving 85% accuracy within 1–5 m range
- Analyzed performance drop beyond 5 m (60%) due to occlusions and extreme viewing angles

Technical Skills

Languages: Python, C++
Computer Vision: OpenCV, camera calibration, image processing, Semantic segmentation, Object detection
Deep Learning: PyTorch, model training, evaluation, loss tuning
Deployment: ONNX, real-time inference optimization
Data: synthetic data generation (GANs, CycleGAN, VAE), dataset curation, Evaluation (FID, KID, LPIPS)
Tools: Git, Linux

Education

M.Sc. Mechatronics **Oct 2023 – Jan 2026**
University of Applied Sciences Ravensburg-Weingarten | ***Grade:*** *1.9* *Weingarten, Germany*

B.Eng. Mechanical Engineering **Jul 2018 – May 2022**
MS University of Baroda | ***Grade:*** *1.2* *Vadodara, India*

Languages

German (B1) — **English** (C1) — **Hindi** (Native)